Certified Algorithms for Combinatorial Optimization Problems

Martin Sokolov

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Contents

1 Introduction

- Introduce the problem FVS in the most general setting (important problem, many ways to solve it)
- Mention approximation algorithms as a way to deal with the NP-hardness of the problem
- Baker's technique as a means of acquiring a PTAS
- Other means of acquiring an approximation ratio, specifically a constant-factor approximation, LP
- BWCA, certified algorithms

1.1 Motivation

1.2 Related Work

This work is inspired by Baker's seminal paper on designing PTASs in planar graphs [1]. There have been other techniques that generalize this approach, namely Demaine and Hajiaghayi [2]'s bidimensionality theory.

The notion of Bilu-Linial stability has given rise to much exploration of structured instances in many problems. In their paper [3] they offer the first evidence that stable instances are much easier than worst-case instances. They propose an exact algorithm for $O(n)$-stable instances of Max Cut. This result was improved by Bilu et al. [4] who designed a $O(\sqrt{n})$ for the same problem. Further research yielded a general approach to analysing stable instances in Makarychev et al. [5]. They applied that approach to graph partitioning problems and designed $O(\sqrt{\log n} \log \log n)$-stable instances of Max Cut and 4-stable instances of Minimum Multiway Cut. The study of perturbation resilience of clustering problems is explored in [6, 7, 8]. On the Travelling Salesman Problem, Mihalák et al. [9] gave an algorithm for 1.8-stable instances. We refer the reader to the survey [10].

On the subject of Beyond the Worst-Case Analysis we refer to Makarychev and Makarychev [11]. In that paper certified algorithms are first described in an attempt to bridge the gap between the worst-case and structured instances. Since then, researchers have made use of the definitions and results in that paper and applied them to other problems. Angelidakis et al. [12] explore the notion of stability and initiate the study of certified algorithms for the Maximum Independent Set problem (MIS). They provide robust and certified algorithms for $(1+\epsilon)$-stable instances of planar MIS, for any fixed $\epsilon > 0$. Covering and Dominating in planar graphs was further explored by Bumpus et al. [13]. They provide $(1+\epsilon)$-certified algorithms for Dominating Set and H-Subgraph-Free-Deletion, which for any ϵ run in time $f(1/\epsilon) \times n^{O(1)}$ on minor-closed classes of graphs of bounded local treewidth.

2 Preliminaries

In this section, we provide the reader with the necessary information in order to follow the paper. We start off with a reminder on Graph Theory. After that, we move to more advanced definitions in the field of Combinatorial Optimization. Finally, we get on topic by introducing perturbation resilience and certified algorithms from the field of Beyond the Worst-Case Analysis of Algorithms.

2.1 Graph Theory

In this section, the authors refer to Bondy and Murty [14]. A *graph* G is an ordered pair denoted as $(V(G), E(G))$. The set $V(G)$ is called a set of *vertices* and the set $E(G)$, a set of *edges*. The incidence function ψ_G associates each edge with an unordered pair of (not necessarily unique) vertices. The ordered pair and the incidence function define the graph G. We are going to use n as the cardinality of the set V and m for the set of edges E. With each edge e of G, let there be associated a real number $w(e)$, called its *weight*. Then G, together with the weighting function $w : E \to \mathbb{R}$, is called a *weighted graph*, and denoted (G, w).

A graph is can be *embedded in the plane*, or is *planar* if it can be drawn on the plane in a way that edges intersect each other only at their ends. Such a drawing is called a *planar embedding* of the graph. A graph G is called *outerplanar* if it has a planar embedding $\widetilde{G}$ in which all the vertices lie on the boundary of the outer face. This is also called a *1-outerplanar graph*. Two forbidden minors of that class are the K_4 and $K_{2,3}$ graphs. In the next subsection we familiarize the reader with Baker's approach [1]. Here we define a *k-outerplanar* to be a graph that has a planar embedding in which the vertices belong to at most k concentric layers. Every planar graph is *k-outerplanar* for some k. Given a graph G, we look for *k-outerplanar* embedding of G for which k is minimal.

2.2 Combinatorial Optimization

In this section, we refer to the book on Combinatorial Optimization by Korte and Vygen [15] and Williamson and Shmoys [16]. Some definitions can be traced to the end of the book by Cygan et al. [17]. We are going to be looking at a couple of problems, namely the Feedback Vertex Set, the

Cycle Packing problem and Travelling Salesman Problem. All of those problems belong to the class of **NP-complete** problems. The former was also part of Karp's original 21 problems [18]. The decision variant of the FVS problem can be defined as:

Definition 2.1: Feedback Vertex Set

Input: A graph G and an integer k.
Question: Does there exist a set X of at most k vertices of G such that $G - X$ is a forest?

The weighted optimization version of the problem can be defined as:

Definition 2.2: Minimum Weight Feedback Vertex Set

Instance:
- An undirected weighted graph (G, w), where $G = (V, E)$ and $w : V \to \mathbb{R}$.

Task: Find a vertex set $X \subseteq V$ such that:
- $G \setminus X$ is acyclic (i.e., a forest)
- The sum of the weights of the vertices in X is minimized.

In some sense dual to the FVS is the Cycle Packing problem. We use the definition from the blue book of Cygan et al. [17].

Definition 2.3: Cycle Packing

Input: A graph G and an integer k.
Question: Does there exist in G a family of k pairwise vertex-disjoint cycles?

The other problem we inspect is an edge-optimization problem, namely the Travelling Salesman Problem (TSP).

Definition 2.4: Travelling Salesman Problem

Instance:
- A set of cities $C = \{c_1, c_2, \ldots, c_n\}$.
- A distance function $w : C \times C \to \mathbb{R}$ that gives the distance between each pair of cities.

Task: Find a permutation π of the cities such that:
- The total distance travelled is minimized, i.e., minimize

$$\sum_{i=1}^{n-1} w(c_{\pi(i)}, c_{\pi(i+1)}) + w(c_{\pi(n)}, c_{\pi(1)}).$$

We define the problems FVS-BCL, MWFVS-BCL and Cycle Packing-BCL to be the problems that cover and pack cycles respectively of bounded length only in the graph. We give only the first definition:

Definition 2.5: FVS-BCL

Input: A graph G, integer k and an integer r.
Question: Does there exist a set X of at most k vertices such that every cycle of length at most r contains a vertex in X?

As we mentioned, all three of the problems are **NP-complete**, so we turn to *approximation algorithms* in order to cope with the hardness of the problems. Let Π be an optimization problem with nonnegative weights and $k \geq 1$. A *k-factor approximation algorithm* for Π is a polynomial-time algorithm A for Π, such that

$$\frac{1}{k} \times OPT(I) \leq A(I) \leq k \times OPT(I)$$

for all instances I of Π. The first inequality shown is true for maximization problems, while the second one holds for minimization problems. We say A has performance ratio k. If a problem falls into that category we say that it is **APX-complete**.

In the last paragraph we mentioned the class of problems **APX** with regards to approximation algorithms. Now we turn to another class of approximation algorithms. **PTAS** is a class of algorithms that works in polynomial time and gets close to an optimal solution with a trade-off.

Definition 2.6: Polynomial-Time Approximation Scheme (PTAS)

A polynomial-time approximation scheme (**PTAS**) is a family of algorithms A_ϵ where there is an algorithm for each $\epsilon > 0$, such that A_ϵ is a $(1 + \epsilon)$-approximation algorithm (for minimization problems). The running time of the algorithm A_ϵ is polynomial, but can depend arbitrarily on $1/\epsilon$. This dependence could be exponential or even worse.

Every problem with a **PTAS** is also in **APX**. The relation between these two classes is similar to the one between **P** and **NP**. For more on the topic, such as *APX-completeness*, *L-reductions* and the *PCP-theorem* we refer to Korte and Vygen [15].

2.2.1 Baker's approach

In this subsection, we are going to describe the Shifting technique. It is described in Baker's seminal paper [1]. What she describes is a general algorithm that partitions the graph into a disjoint union of levels of vertices. By analysing slices of these levels and more specifically the overlap between these slices, one obtains an approximation ratio for the solution of the optimization problem. The slices constitute a number of levels that are specifically chosen for a problem and it should hold that an exact solution can be found on one slice. It must also hold that this solution can be found in optimal time.

There is also another way to view this approach. Instead of analysing the overlap between slices, one could partition the graph by removing said overlap. This way we get a disjoint union which does not sum up to the total solution, but we can argue that we did not stray away so much from the optimum solution. This approach works for hereditary optimization problems like Independent Set, Max-Cut, Maximum P-Matching, etc.

We are going to give two examples. The first is the general shifting technique and the example problem for it will be Vertex Cover. Then we will explore the second perspective and take Independent Set as an example. It is important to note that the general technique is much more powerful than partitioning by means of removal of vertices. In the case of Vertex Cover we can still use the second approach and come to a PTAS, but the same does not hold for Dominating Set. However, using the general Shifting technique we can obtain a PTAS for Dominating Set in planar graphs. This was done by Marzban and Gu [19]. They designed a PTAS with a modified approximation ratio of $(1+2/k)$ in $O(2^k \times n)$ time. The two in the approximation ratio appears, because of the overlap needed for Dominating Set between slices which is two levels.

What follows is a brief explanation of Baker's Shifting technique with Vertex Cover as an example.

Algorithm 1 Baker's Shifting technique

Require: Planar graph $G = (V, E)$, parameter k
Ensure: $(1 + \epsilon)$-approximation for Vertex Cover

1: Perform BFS from some arbitrary vertex r
   i: *shift*; j: *slice*
2: **for** each $i = 0$ to $k - 1$ **do**
3: Let $G_{i,j}$ be the subgraph induced on vertices at levels $j \times k + i$ through $(j + 1) \times k + i$ for $0 \le i < k$.
4: $\forall i, j$ compute $S_{i,j}$, the min. VC of $G_{i,j}$.
5: $\forall i$ let $S_i = \bigcup_j S_{i,j}$
6: **end for**
7: S_i with the minimum cardinality

Analysis of the algorithm:

Let V_i be the set of vertices at levels $i \mod k$:

$$V_i = \{v \in V \mid \text{level}(v) \equiv i \mod k\}$$

Let OPT_i denote $OPT \cap V_i$. We notice that OPT_i are disjoint and $OPT = \bigcup_i OPT_i$. Therefore we can make the following argument:

1. $\exists q : |OPT_q| \le \frac{1}{k} \times OPT = \epsilon |OPT|$
2. $S_q \le \sum_j S_{qj}$, for some shift q we have this inequality, because in step 5. of the algorithm we define the set to be a union over all slices and some vertices may appear twice in the boundary layers.
3. $\sum_j S_{qj} \le \sum_j |OPT \cap V(G_{qj})|$, we know that S_{qj} is a min. VC for G_{qj}, while the optimal solution may choose different vertices, especially on the boundary, for the global optimality of the solution.

4. $\sum_j |OPT \cap V(G_{qj})| \leq |OPT| + |OPT_q|$, if we consider the graphs $\{G_{qj}\}$, then we can argue that each vertex appears once, except those at levels $q \mod k$, which appear twice.

From the first and fourth step we can conclude that

$$S_q \leq |OPT| + |OPT_q| \leq (1 + \epsilon) \times |OPT|$$

Now we take a look at the second approach we described.

Algorithm 2 Baker's technique with removal

Require: Planar graph $G = (V, E)$, parameter k
Ensure: $(1 - \epsilon)$-approximation for MIS

```
Perform BFS from some arbitrary vertex r
Label each vertex according to its layer i mod k.
for each i = 0 to k − 1 do
    Remove the i-th layer and solve MIS optimally on the remaining graph
    Combine the solution with the i-th layer to form a feasible solution for
    the original graph
end for
Return the best solution found among all iterations
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Consequence of the algorithm:

1. for $k = 1/\epsilon$ compute partition $P_1, \ldots, P_k$.
2. for some q $(1 \leq q \leq k)$, $|OPT \cap P_q| \leq \frac{1}{k} \times |OPT| = \epsilon \times |OPT|$, because of the pigeonhole principle
3. hence, $|OPT \cap (G - P_q)| \geq (1 - 1/k) \times |OPT| = (1 - \epsilon) \times |OPT|$
4. return $max_i|\text{MIS}(G - P_i)|$

In step 3. of the consequences of the algorithm we skipped much of the analysis that we did for the shifting technique. What we argue however, is that for this particular problem for some P_q, a small fraction of the solution resides within that partition. Therefore, the problem allows to forget about that part and the rest of the solution will still be an independent set of bounded size.

The first approach is more general and powerful for constructing algorithms, but it doesn't apply well to the Dominating Set problem. The key issue is that the removal technique relies on intersecting an optimal solution with a subgraph, expecting it to remain valid for the subgraph. This fails for Dominating Set, since a vertex in the subgraph could still be dominated by a vertex outside the subgraph.

It is shown that the general shifting technique can be used to obtain a PTAS for Dominating Set. The interesting part is that in order to bound the solution one must make the slices overlap on two levels. We refer to Marzban and Gu [19] for the explanation as to why the last is true and how to obtain the $(1 + 2k)$ approximation ratio.

2.2.2 Linear Programming and Randomized Rounding

One way of approximating solutions is done through Linear Programming (LP). For a more detailed review on Linear Programming the reader can consult Chapter 3 of Korte and Vygen [15]. The algorithm we use solves a linear relaxation which provides fractional values as solutions. We then round this fractional solution to an integral one, but rather than doing it deterministically, we do the process randomly. This technique is called *randomized rounding.*

2.3 Beyond the Worst-Case Analysis of Algorithms

In this section we explore perturbation-resilience, also known as γ-*stability* or Bilu-Linial stability. The authors refer to the original paper of the two authors Bilu and Linial [3] as well as a survey on the matter by Makarychev and Makarychev [10]. In a later work Makarychev and Makarychev define γ-*certified algorithms* [11]. In the final paragraph we take a look back at combinatorial optimization problems through the lens of constraint satisfaction.

The following definitions for γ-perturbation are for edge-optimization and vertex-optimization problems respectively. Afterwards, we are going to define stability and certified algorithms for vertex-optimization problems only.

Definition 2.7: γ-perturbation for Edge-Optimization problems

Consider a weighted graph $G = (V, E, w)$ with a set of edge weights w_e. An instance $G' = (V, E, w')$ is a γ-perturbation ($\gamma \in \mathbb{R}_{\geq 1}$) of G if $w(e) \leq w'(e) \leq \gamma \times w(e)$; that is, if we can obtain a perturbed instance from the original by multiplying the weight of each edge by a number between 1 and γ (the number may vary for each edge).

From now on, the definitions will concentrate on Vertex-Optimization problems.

Definition 2.8: γ-perturbation for Vertex-Optimization problems

Consider a weighted graph (G, w). For any $\gamma \in \mathbb{R}_{\geq 1}$ a γ-perturbation of the weight function $w : V \to \mathbb{R}$ is a function $w' : V \to R$ satisfying $w(v) \leq w'(v) \leq \gamma \times w(v)\ \forall v \in V$.

Definition 2.9: γ-stability

For any $\gamma \in \mathbb{R}_{\geq 1}$, a γ-stable instance (G, w) of a *vertex-minimization* problem Π is an instance admitting a unique optimal solution S which remains optimal even under γ-perturbations of (G, w).

Definition 2.10: Certified Algorithm

A γ-certified solution to an instance (G, w) of a weighted vertex-optimization problem Π is a pair (S, w') where w' is a γ-perturbation of w and S is an optimal solution on (G, w').

Definition 2.11: Combinatorial Optimization problems (Constraint Satisfaction)

An instance of a *combinatorial optimization* problem is specified by a set of feasible solutions $\mathcal{S}$ (the solution space) a set of constraints $\mathcal{C}$ and constraint weights $w_c > 0$ for $c \in \mathcal{C}$. Typically the solution space is exponential in size (as is the case with MWFVS with the number of cycles in a graph) and is not provided explicitly. We say that a feasible solution $s \in \mathcal{S}$ satisfies a constraint $c \in \mathcal{C}$ if $c(s) = 1$.
Consider the minimization and maximization objectives.

- The minimization objective is to minimize the total weight of the unsatisfied constraints: find $s \in \mathcal{S}$ that minimizes $\sum_{c \in C} w_c \times (1 - c(s)) = w(\mathcal{C}) - val_{\mathcal{I}}(s)$, where $w(\mathcal{C}) = \sum_{c \in \mathcal{C}} w(c)$ is the total weight of the constraints.
- The maximization objective is to maximize the total weight of the satisfied constraints: find $s \in \mathcal{S}$ that maximizes $val_{\mathcal{I}}(s) = \sum_{c \in \mathcal{C}} w(c)$.

3 Research Question

In this work, we explore several questions. The first two are of most significance, while the problems that follow will be tackled in what time is left.

- Explore the FVS-BCL problem in planar graphs.
  - To that end we are going to make use of Baker's shifting technique [1] as presented in Algorithm 1.
  - By using the modified version with an overlap between layers of $\lfloor \frac{r}{2} \rfloor + 1$, we capture the essence of breaking cycles of bounded length r.
  - Baker's technique is known to work for problems that involve local conditions on nodes and edges and our imposed restrictions on the FVS allow us to deploy this framework.
  - We extend this method to devise a PTAS for the weighted version of the problem, MWFVS-BCL for planar graphs.
- Initiate the study of certified algorithms for MWFVS-BCL. We provide robust and certified algorithms for $(1 + \epsilon)$-stable instances of planar MWFVS-BCL, for any fixed $\epsilon > 0$.
- Future Directions:
  - Investigating the Cycle Packing-BCL problem. As a hereditary maximization task, we make use of the alternative approach to Baker's technique, namely Algorithm 2 with removal.
  - Exploring the translation of PTASs for the TSP to $(1 + \epsilon)$-certified algorithms.

Firstly, we are going to explore the problem of FVS in planar graphs where we try to break all cycles of bounded length r. To that end we are going to make use of Baker's shifting technique [1]. The reader can imagine a modified version of Algorithm 1 where the overlap between layers is $\left\lfloor \frac{r}{2} \right\rfloor + 1$. On an intuitive level, this number captures the fact that each (bounded) cycle, that has to broken, starting from a node in the BFS tree *has to go back to the starting vertex*. Baker's technique is known to work for problems that involve local conditions on nodes and edges and our imposed restrictions on the FVS allow us to deploy this framework. Then we move on to the weighted version of the problem. For the same conditions we devise a PTAS for the MWFVS problem.

Secondly, we initiate the study of certified algorithms for MWFVS. More precisely, we provide robust and certified algorithms for $(1 + \epsilon)$-stable instances of planar MWFVS with cycles of bounded length, for any fixed $\epsilon > 0$.

The authors plan to continue this area of research if time allows it, by exploring the following avenues.

The first one is looking at the Cycle Packing problem. In the Preliminaries section we described hereditary maximization tasks. This problem falls into that category, thus allowing us to make use of the alternative approach to Baker's technique, namely Algorithm 2 with removal.

The second one involves checking if we can translate current PTASs for the TSP to $(1 + \epsilon)$-certified algorithms.

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